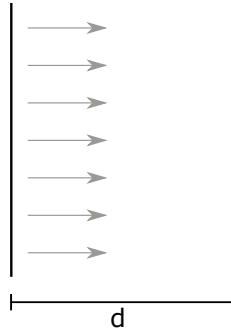


Instructions:

- Solutions are due by 9:00 a.m. the next morning.
- While points may differ between Problem Sets each Problem Set will carry equal weight for course grade.
- Please list your collaborators on your submission.

**Problem 1.** Relativistic Diode 20 points

Calculate the current density  $J$  for an extreme relativistic 1D diode. (i.e. ignore the small region of the diode for which  $\beta \simeq 1$  is not a good approximation.) Assume space-charge limited emission. Let the diode have length  $d$ , and voltage  $V$ , and the ion species have mass  $m$  and charge  $q$ . Sketch  $\log J$  vs.  $\log V$  for both a non-relativistic diode and an extreme relativistic diode and comment on how the two cases scale.



Hint:

$$\nabla \cdot E = \frac{\rho}{\epsilon} \quad \text{and} \quad \frac{\partial \rho}{\partial t} + \nabla \cdot J = 0$$

are both relativistically correct equations.

**Solution: Relativistic Diode 20 points.**

As stated in the problem the continuity equation is relativistically correct:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot J = 0$$

$\Rightarrow$  As in the non-relativistic case  $J$  is constant throughout the gap. For a true 1D diode there can be no transverse magnetic field. Poisson's equation remains valid,

$$\nabla \cdot E = \frac{\rho}{\epsilon}$$

So  $dE/dz = \rho/\epsilon_0$  and  $J = \rho v$ . In the extreme relativistic case  $v \cong c$ . So  $\rho \cong J/c \cong \text{constant}$  through the gap.

$$dE/dz = J/\epsilon_0 c$$

$$\Rightarrow E = (J/\epsilon_0 c)z + \text{const.},$$

but  $E = 0$  at  $z = 0$  (space-charge limited emission) so the constant = 0.

$$\Phi = \int E dz \Rightarrow \phi = (J/\epsilon_0 c)z^2/2.$$

Here we use  $\Phi = -\phi$ , where  $\phi$  = the actual potential.

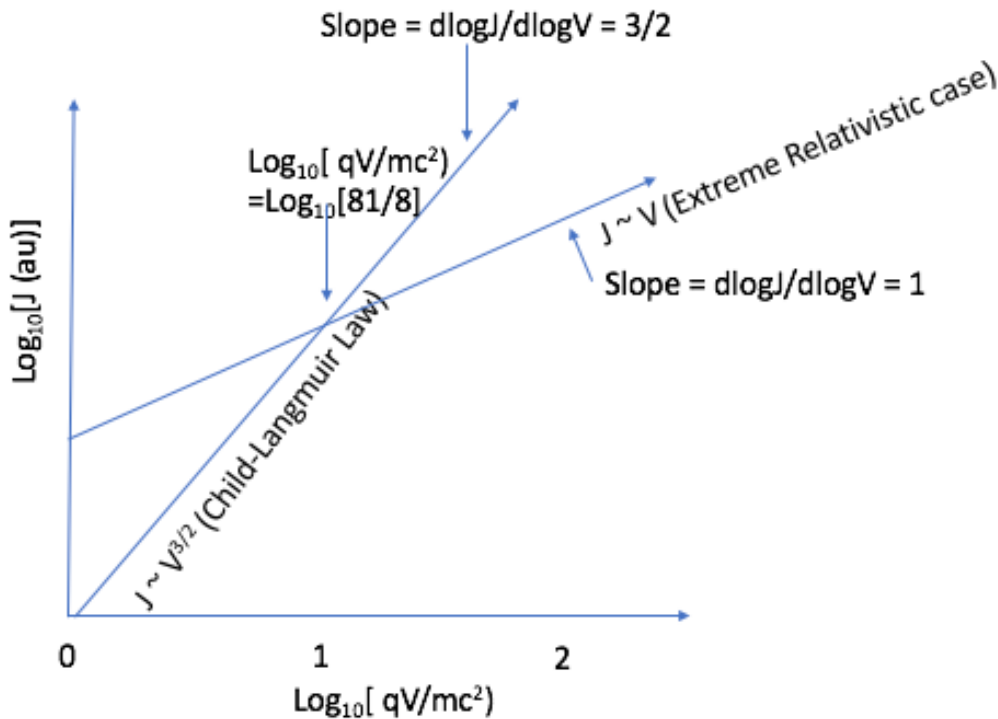
At  $z = d$ ,  $V = V_0$ .

$$\Rightarrow V = Jd^2/(\epsilon_0 c^2) \quad \text{or} \quad J = 2\epsilon_0 c V/d^2.$$

The Child-Langmuir Law is:

$$J = (4\epsilon_0/9)(2q/m)^{1/2}V^{3/2}/d^2.$$

We see that  $d \log J / d \log V = 3/2$  for the non-relativistic case, and  $d \log J / d \log V = 1$  for the relativistic case.



**Problem 2.** FRIB Solenoids 15 pts.

FRIB will use superconducting solenoids for beam focusing in the three linac segments. Consider  $U_{238}^{+78}$  (post stripper) accelerated to a kinetic energy of 128 MeV/u. The solenoids have an axial length of approximately 0.5 m and a field strength of 7 T.

- a) 5 pts: Estimate  $[B\rho]$  for the ion in T-m.
- b) 5 pts: Estimate the focal length,  $f$ , of the solenoid assuming  $B_z \approx 7$  T over the axial length and is zero outside (hard edge solenoid). Is the thin lens approximation reasonable? Why?
- c) 5 pts: Suppose superconducting cable with an average current density of  $J$  is wound to a thickness  $T$  around the beam pipe. Derive a formula to roughly estimate the central field,  $B_z$  in terms of  $J$  and  $T$ . If the coil thickness is  $T = 2$  cm what is the required current density in A/mm<sup>2</sup> to produce  $B_z = 7$  T?

FRIB Solenoids 15 pts.

$$^{238}\text{U}^{78+}, W = 128 \text{ MeV/u}, \ell = 0.5 \text{ m}, B_0(0) = 7 \text{ T}.$$

- a) 5 pts:

$$[B\rho] = \frac{p}{q} = \frac{\sqrt{(E/c)^2 - (mc)^2}}{q} = \frac{m\beta\gamma c}{q} \quad E = \mathcal{E} + mc^2$$

$$E \simeq 252158 \text{ MeV}, mc^2 \simeq 221695 \text{ MeV} \Rightarrow p \simeq 120 \text{ GeV/c. } q = 78e. \Rightarrow [B\rho] \simeq 5.1 \text{ T-m.}$$

- b) 5 pts: Using the results from the class notes on a thin lens solenoid:

$$\frac{1}{f} = \left( \frac{B_0(0)}{2(B\rho)} \right)^2 \ell = \left( \frac{7 \text{ T}}{2(5.1 \text{ Tm})} \right)^2 (0.5 \text{ m}) \simeq 0.2355 \text{ m}^{-1}$$

$$\Rightarrow f \simeq 4.25 \text{ m}, \frac{f}{\ell} \simeq 8.5$$

The thin lens approximation may be reasonable because the focal length is much longer than the length of the solenoid.

- c) 5 pts: Take a square Amperian loop with side length  $\ell > T$  located on the wire such that one side is inside the coil and parallel to the magnetic field and the opposite edge is outside the coil.

$$B_z \ell = \mu_0 J T \ell$$

$$\Rightarrow J = \frac{B_z}{\mu_0 T}$$

$$T = 2 \text{ cm}, B = 7 \text{ T}, \ell = 0.5 \text{ m} \Rightarrow J = 279 \text{ A/mm}^2.$$

### Problem 3. Magnetic chicane

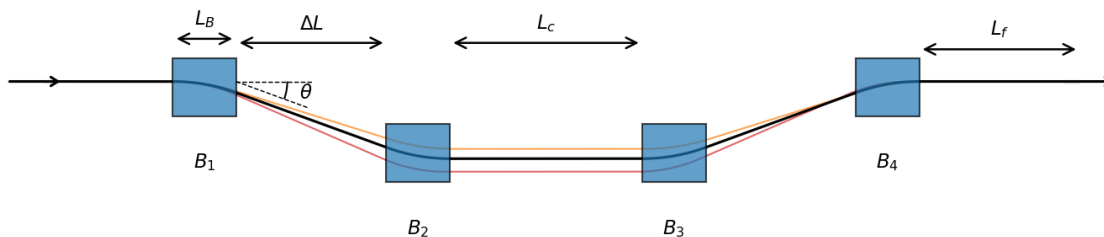


Figure 1: Magnetic chicane

Bunch compression of an electron beam is achieved by making leading electrons propagate slower and trailing electrons propagate faster, hence shortening the bunch length. In the relativistic regime where electrons travel in the speed of approaching the speed of light, energy-dependent path-length-difference is induced to make leading (trailing) electrons travel a longer (shorter) path to achieve bunch compression. A magnetic chicane is a type of bunch compressor that employs four dipole magnets arranged in the form shown in Fig. 1.

- a) 1 pts: The straight-through path (if no dipoles were present) has length:

$$s_{\text{straight}} = 4L_B + 2\Delta L + L_c + L_f .$$

The path travelled by an electron beam passing through a magnetic chicane is shown as the black trajectory line shown in Fig. 1. Find the expression of path length  $s(\theta)$ .

- b) 5 pts: Find the path-length difference,  $\Delta s = s(\theta) - s_{\text{straight}}$ , do Taylor expansion up to the second order and show that

$$\Delta s = s(\theta) - s_{\text{straight}} \approx \theta^2 \left( \Delta L + \frac{2}{3} L_B \right) . \quad (1)$$

- c) 6 pts: The path length difference can be expressed as

$$\Delta s = \Delta s_0 + R_{56}\delta + T_{566}\delta^2 + U_{5666}\delta^3 + \mathcal{O}(\delta^4) , \quad (2)$$

where  $\Delta s_0$  is the path-length difference of the designed energy  $E_0$ . Plug in  $\theta = \theta_0/(1 + \delta)$  into eq. 1, then expand it up to the third order, then compare it with eq. 2 to show that

$$R_{56} \approx -2\theta_0^2 \left( \Delta L + \frac{2}{3} L_B \right) ,$$

$$T_{566} \approx -\frac{3}{2} R_{56} ,$$

$$U_{5666} \approx 2R_{56} .$$

- d) 3 pts: Check Fig. 2. This FEL has chicane bunch compressors that compress beams from

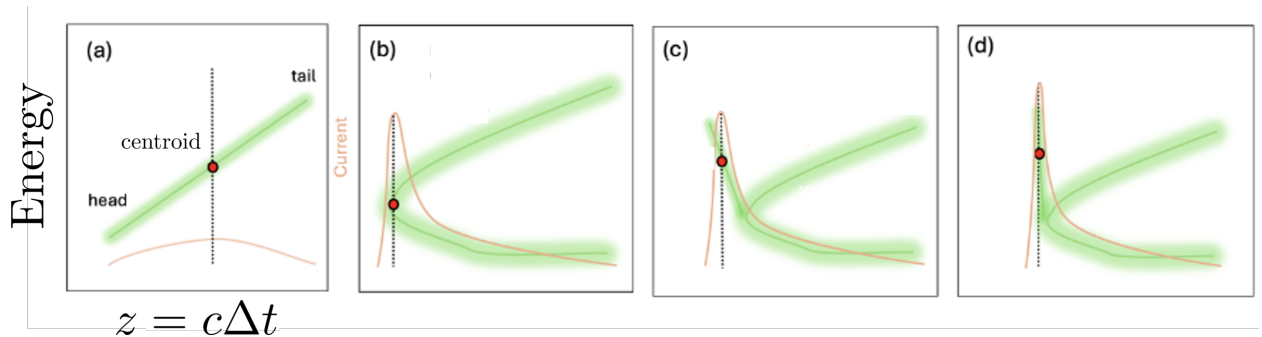


Figure 2: Longitudinal phase space at (a) before chicane, (b) after chicane, (c) before a special bunch compressor, (d) final outcome.

(a) to (b). Beams get worse from (b) to (c) due to wakefields. After that, a special bunch compressor compress beams from (c) to (d). Does this special bunch compressor need  $R_{56} > 0$  or  $R_{56} < 0$ ? Why?

a) 1 pts:

- **Inside each dipole:** The arc length is  $L_B \theta / \sin \theta$  (since the projected length along the beam axis is  $L_B = \rho \sin \theta$  where  $\rho$  is the bending radius, so the arc  $= \rho \theta = L_B \theta / \sin \theta$ ). There are 4 dipoles.
- **Drifts between dipoles 1–2 and 3–4:** Each drift of projected length  $\Delta L$  becomes  $\Delta L / \cos \theta$  along the particle trajectory. There are 2 such drifts.
- **Central drift  $L_c$ :** The particle returns to the axis before the central drift (by symmetry of the first half), so this contributes  $L_c$ .
- **Field-free region  $L_f$ :** Contributes  $L_f$  (particle is on-axis).

The total path length is:

$$s(\theta) = \frac{4L_B \theta}{\sin \theta} + \frac{2\Delta L}{\cos \theta} + L_c + L_f .$$

b) 5 pts: For small  $\theta$ :

$$\frac{\theta}{\sin \theta} \approx 1 + \frac{\theta^2}{6} , \quad \frac{1}{\cos \theta} \approx 1 + \frac{\theta^2}{2} .$$

So:

$$s(\theta) \approx 4L_B \left(1 + \frac{\theta^2}{6}\right) + 2\Delta L \left(1 + \frac{\theta^2}{2}\right) + L_c + L_f .$$

Then:

$$\Delta s = s(\theta) - s_{\text{straight}} \approx \theta^2 \left( \Delta L + \frac{2}{3} L_B \right) .$$

c) 6 pts:

$$\begin{aligned} \Delta s &= \theta^2 \left( \Delta L + \frac{2}{3} L_B \right) , \\ &\approx \left( \frac{\theta_0}{1 + \delta} \right)^2 \left( \Delta L + \frac{2}{3} L_B \right) , \\ &\approx \theta_0^2 \left( \Delta L + \frac{2}{3} L_B \right) (1 - 2\delta + 3\delta^2 - 4\delta^3 + \mathcal{O}(\delta^4)) , \\ &= \Delta s_0 + R_{56} \delta + T_{566} \delta^2 + U_{5666} \delta^3 + \mathcal{O}(\delta^4) , \\ \therefore R_{56} &\approx -2\theta_0^2 \left( \Delta L + \frac{2}{3} L_B \right) , \\ T_{566} &\approx -\frac{3}{2} R_{56} , \\ U_{5666} &\approx 2R_{56} . \end{aligned}$$

d) 3 pts: This bunch compressor needs  $R_{56} > 0$ .

Explanation 1: At (c), the part of beam with higher energies than the beam centroid (red dot) has a positive chirp.  $R_{56} > 0$  decompress that part of beam and push it towards the current spike. The part of beam with a negative slope will be compressed with  $R_{56} > 0$ , and hence being pushed towards the current spike.

Explanation 2: With  $R_{56} > 0$ ,

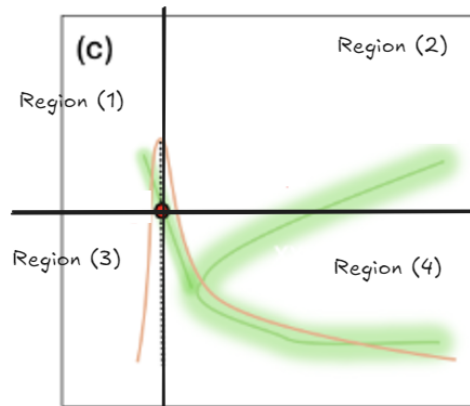


Figure 3

- i. Higher energy electrons travel a longer path.
- ii. Lower energy electrons travel a shorter path.

Therefore, in Fig. 3,

- i. Higher energy electrons in region (1) move towards the centroid (red dot).
- ii. Higher energy electrons in region (2) move away from the centroid.
- iii. Lower energy electrons in region (3) move away from the centroid.
- iv. Lower energy electrons in region (4) move towards the centroid.

Please check Yasutome et al IPAC 2026 proceeding paper for reference.